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Oxyfluoride glasses containing LiNbO_3

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Abstract

The MoO_3 – BaF_2 – NaF – ZnF_2 – $\text{NbO}_{2.5}$ – LiNbO_3 and NbO_2F – BaF_2 – LiF – LiNbO_3 oxyfluoride glass systems were investigated. Glasses were prepared by fusion at 850°C for 30 min in a platinum crucible and cast in a brass mould at 200°C. For each system, the effect of the LiNbO_3 content (to 15 mol%) on the physical properties of the samples, such as glass transition (T_g) and crystallisation (T_x) temperatures, densities and refractive indices were measured. Density (about 4 g cm^{−3}), and refractive index ($n > 1.7$) were measured as a function of dopant concentration and glass processing. Optical transmission measurements in the IR range were also carried out and samples were transparent up to 5 μm . Thermal analysis by differential scanning calorimetry was used to determine the characteristic temperatures of prepared glasses and to define the stability factor ($\Delta T = T_x - T_g$). The glass transition temperature of these glasses is about 250°C, with ΔT around 80°C. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

LiNbO_3 (LN) is a ferroelectric material that is potentially applicable in optoelectronics, acousto-optics and piezoelectric devices [1]. However, the Czochralski method used for single crystal growth is time-consuming and expensive and it is, therefore, important to investigate alternative processes less expensive and easier to prepare LiNbO_3 (LN) or even a composite material containing it [2].

Oxyfluoride glasses have some physical properties that are quite similar to those of LiNbO_3 . These glasses have a transmission window from 400 nm to 5 μm , similar to LN. The refractive indices are greater than 1.7 and, since LN has a refractive index of 2.28, scattering problems at the glass-LN interface will be reduced. The objective

of this work is to add LiNbO_3 particles to a glassy matrix and obtain a stable glass. Two kinds of oxyfluoride glasses were tested, a molybdenum based glass (30 MoO_3 –20 BaF_2 –30 NaF –20 ZnF_2) previously reported by Hager and Poulain [3] and a niobium based glass (36 NbO_2F –40 BaF_2 –24 LiF) studied by Lipeng [4].

2. Experimental procedure

Glass samples, whose molar compositions are listed in Table 1, were prepared by melting and casting. The chemical compounds were weighed and mixed in appropriate proportions. The LiF , NaF , BaF_2 , ZnF_2 fluoride compounds were supplied by BDH-Merck UK, and MoO_3 , Nb_2O_5 were supplied by Merck and Aldrich, respectively. The mixture was placed in a platinum crucible and an amount of ammonium bifluoride (Aldrich), being 15 wt% of the total mixture, was added. The

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Table 1

Composition (mol%) and physical properties of oxyfluoride glasses

Sample code	MoO ₃	NbO _{2.5}	LiF	NaF	BaF ₂	ZnF ₂	LiNbO ₃	T_g (°C)	T_x (°C)	T_p (°C)	ΔT (°C)	d (g cm ⁻³)
V1	30			30	20	20		227	304	322	77	4.07
V2	25	5		30	20	20		224	306	321	82	4.03
V3	20	10		30	20	20		248	326	355	78	4.15
V4	15	15		30	20	20		267	375	391	108	4.32
V5	10	20		30	20	20		272	355	374	83	4.27
V6	5	25		30	20	20		274	356	367	82	
V7	15	15	5	25	20	20		249	337	377	88	4.35
V8	15	15	10	20	20	20		261	348	378	87	4.36
V9	15	15	15	15	20	20		245	329	336	84	4.36
V10	25			25	20	20	5	241	329	353	88	4.32
V11	20			20	20	20	10	248	337	356	89	4.34
V12	15			15	20	20	15	267	324	334	57	4.37
V13 ^a		31	19		40		5	246	303	313	57	4.40
V14		26	14		40		10	268	356	367	88	4.53
V15		21	9		40		15	313	421	463	118	4.64
density	4.69	4.47	2.63	2.55	4.89	4.95	4.61	temperature error is $\pm 2^\circ\text{C}$; density error is ± 0.01 g cm ⁻³				

^a quenched glass.

material was heated to 850°C for 30 min as a refining stage to improve homogeneity. The melted glass was then poured into a brass mould at 200°C. The resulting glass samples, which were used in this work, were at least 2 mm thick. An annealing treatment was also done at 15°C below the glass transition temperature (T_g) for 2 h to eliminate internal stresses and the cooling rate to room temperature was 1°C min⁻¹. For glasses containing LN, the powdered LN single crystal (particle size <45 µm and average size is 25 µm) was introduced during the refining step at 850°C for 30 min.

Thermal analysis (TA Instruments DSC 2090) for each glass sample was carried out at 10°C min⁻¹. Glass transition (T_g) and onset of crystallisation (T_x) temperatures were determined as the point where the DSC curves depart from the horizontal baselines (Table 1). The sample density was measured by the Archimedes method in CCl₄ as a function of composition. The experimental error was $\pm 2^\circ\text{C}$ for temperatures and ± 0.01 g cm⁻³ for density. Optical transmissions of polished samples (3 mm thick) in the infrared region were measured by spectrophotometry (FTIR Nicolet 850). For glass structure investigation, we made IR vibrational spectra in the 1100–400 cm⁻¹ range on

pellets prepared from a mixture of 200 mg of KBr (Merck 99%) and 10 mg of powdered glass.

3. Results

3.1. Glass compositions and their physical properties

Table 1 summarises all compositions investigated in the present work, as well as their physical properties such as T_g , glass transition; T_x , onset of crystallisation; T_p , maximum crystallisation temperature; glass stability factor, $\Delta T = T_x - T_g$, and density.

The T_g for the V1 type glass in the molybdenum series (V1–V12) was 227°C and the density was 4.07 g cm⁻³. The NbO_{2.5} substitution for MoO₃ in the set (V2–V6) increased T_g and densities. The stablest glass (largest ΔT) was obtained when half of the molybdenum content was replaced by NbO_{2.5} (V4).

Based on these results, and with the purpose of preparing a relatively stable glass containing Li and Nb cations, NaF was replaced by LiF in the V7–V9 series. In this series derived from the V4 composition, we did not observe any changes in T_g , T_x , T_p , or density. Hence, lithium can replace

sodium in these oxyfluoride glasses without inducing any destabilisation. The $T_x - T_g$ was around 85°C.

Increases of T_g s and densities were observed when LiNbO_3 replaced MoO_3 and NaF simultaneously (keeping the cationic concentration constant) in the glass set V10–V12, while the ΔT decreased ($\Delta T = 57^\circ\text{C}$) for concentrations of $\text{LiNbO}_3 \geq 10 \text{ mol}\%$.

In the case of the $\text{LiF-BaF}_2\text{-NbO}_{2.5}$ system (V13–V15), where LiF and $\text{NbO}_{2.5}$ were simultaneously replaced by LN , T_g s, T_x s and densities increased. The most stable glass was obtained with 15 mol% LN with ΔT equal to 118°C. LN incorporation in a vitreous matrix is better in these samples than in the molybdenum based samples.

3.2. Refractive index

The technique used to measure the refractive index is described by Alonso et al. [5]. 4 mm diameter cylindrical rods were prepared and polished. Measurements were made for V14 (10 mol% LN) and V15 (15 mol% LN) samples and the refractive indexes were 1.70 and 1.90 respectively, while the error in measurements was ± 0.05 . The introduction of LN in this glass matrix increased the refractive index, as expected.

3.3. Optical properties

Infrared transmission spectra of the oxyfluoride glass samples (3 mm thick) are shown in Figs. 1 and 2. Maximum transmission in the infrared region was at least 75%. The multiphonon absorption edge is not resolved because it overlaps with an absorption band at 5.5 μm .

No absorption bands in the UV-Visible region were detected in the V14 and V15 samples, indicating that the valence of Nb in the glasses is Nb^{5+} [4]. A slight yellow-brown colour was observed in Mo containing glasses, which we attribute to the presence of Mo in oxidation states less than +6.

Fig. 3 shows infrared spectra absorption bands at ~ 915 , 860, 800, 625, and 475 cm^{-1} from KBr pellets containing V4, V12 and V15 powdered

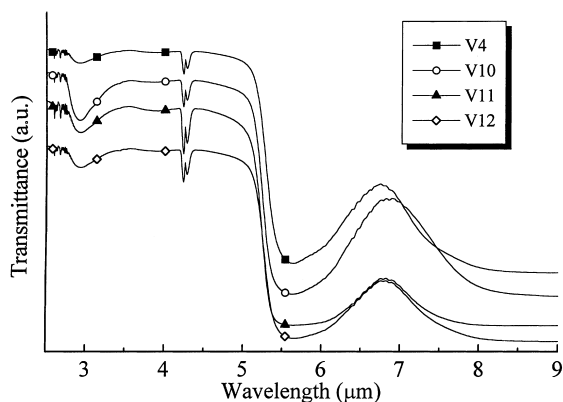


Fig. 1. Transmission spectra of MoO_3 based oxyfluoride glasses containing $\text{NbO}_{2.5}$ and LiNbO_3 .

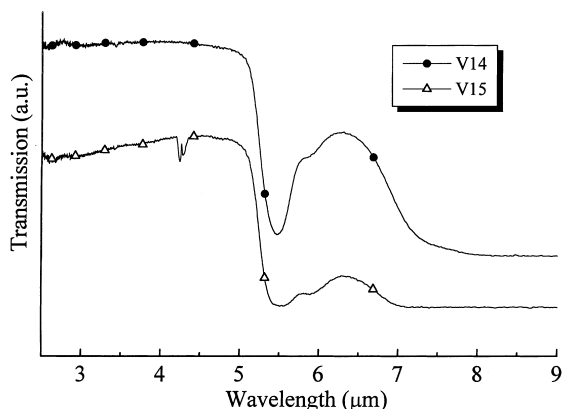


Fig. 2. Transmission spectra of glasses containing LiNbO_3 .

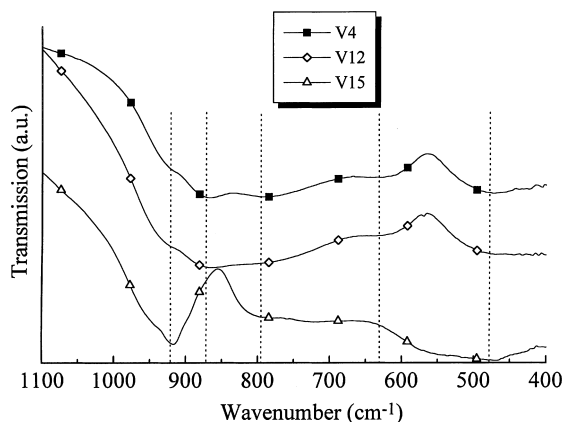


Fig. 3. Infrared spectra of powdered glass in KBr pellets.

glass. Vertical dashed lines on the graph indicate the five regions considered.

4. Discussion

Niobium incorporation in the molybdenum based oxyfluoride glass resulted in a denser material, in spite of having larger ionic radii ($r_{\text{Nb}^{5+}} = 69$ pm, $r_{\text{Mo}^{6+}} = 60$ pm) and lower density ($d_{\text{NbO}_{2.5}} = 4.47 \pm 0.01$ g cm⁻³, $d_{\text{MoO}_3} = 4.69 \pm 0.01$ g cm⁻³) [6]. The introduction of niobium induced a collapse of the glass structure (cf. density increase), leading to an increase of the average cation–anion bonding strength. The macroscopic effect is an increase of T_g .

In the IR transmission spectra (Fig. 1), the band at 5.5 μm is at the wavelength of the first overtone of the ν_3 vibration mode of the MoO_4 tetrahedron [3]. This band may also be attributed to the $\text{Nb}=\text{O}$ bonding in $[\text{NbOF}_5]^{2-}$ [4] in the samples containing $\text{NbO}_{2.5}$ and LN (Fig. 2). The band observed around 3 μm in the V4–V12 set is attributed to hydroxyl ions, incorporated in the glass from ambient moisture during the glass preparation process.

Investigation of the glass structure by IR vibrational spectroscopy showed the presence of five absorption bands (Fig. 3). For crystalline oxyfluoroniobates [7], the bands at ~ 925 cm⁻¹ are ascribed to the vibrations of the $\text{Nb}=\text{O}$ double band. The bands observed at ~ 860 cm⁻¹ and 475 cm⁻¹ are attributed to the edge-share NbO_6 octahedron observed in $\text{H-Nb}_2\text{O}_5$ crystals [8]. Note that in the present work, the band in the 860 cm⁻¹ region appeared only in the spectra of the V4 and V12 samples containing Mo. The band centred in 800 cm⁻¹ is due to the vibration of the $-\text{Nb}-\text{O}-\text{Nb}-\text{O}-$ bridging bonds. The $\text{Nb}-\text{F}$ vibrations are usually observed in the 600–400 cm⁻¹ range [9].

5. Conclusion

These glasses have properties, such as density (~ 4.5 g cm⁻³), refractive index ($n_D > 1.7$) and transparency in the infrared region (5 μm), similar to LN single crystals. Their melting temperature is around 700°C, which is very low when compared to the LN fusion temperature of 1253°C. We conclude that these vitreous oxyfluoride materials are potential candidates for the preparation of glass-ceramics containing LN.

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